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Modern Application Development - 2 Project Report

Quiz Master Application

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Description

Quiz Master - V2 is an online exam preparation platform that allows users to take quizzes across multiple subjects and chapters. The platform supports role-based access, with an admin responsible for managing subjects, chapters, quizzes, and users, while registered users can attempt quizzes and track their scores.

The application ensures structured learning by organizing content into subjects → chapters → quizzes → MCQ-based questions. Each quiz is timed, and users receive instant scoring with performance tracking.

Additional features include scheduled notifications for inactive users, monthly reports summarizing quiz activity, a. The system is optimized with Redis caching for fast API responses and background jobs using Celery. The frontend is built using Python and VueJS, while SQLAlchemy ORM manages the database.

Technologies Used:

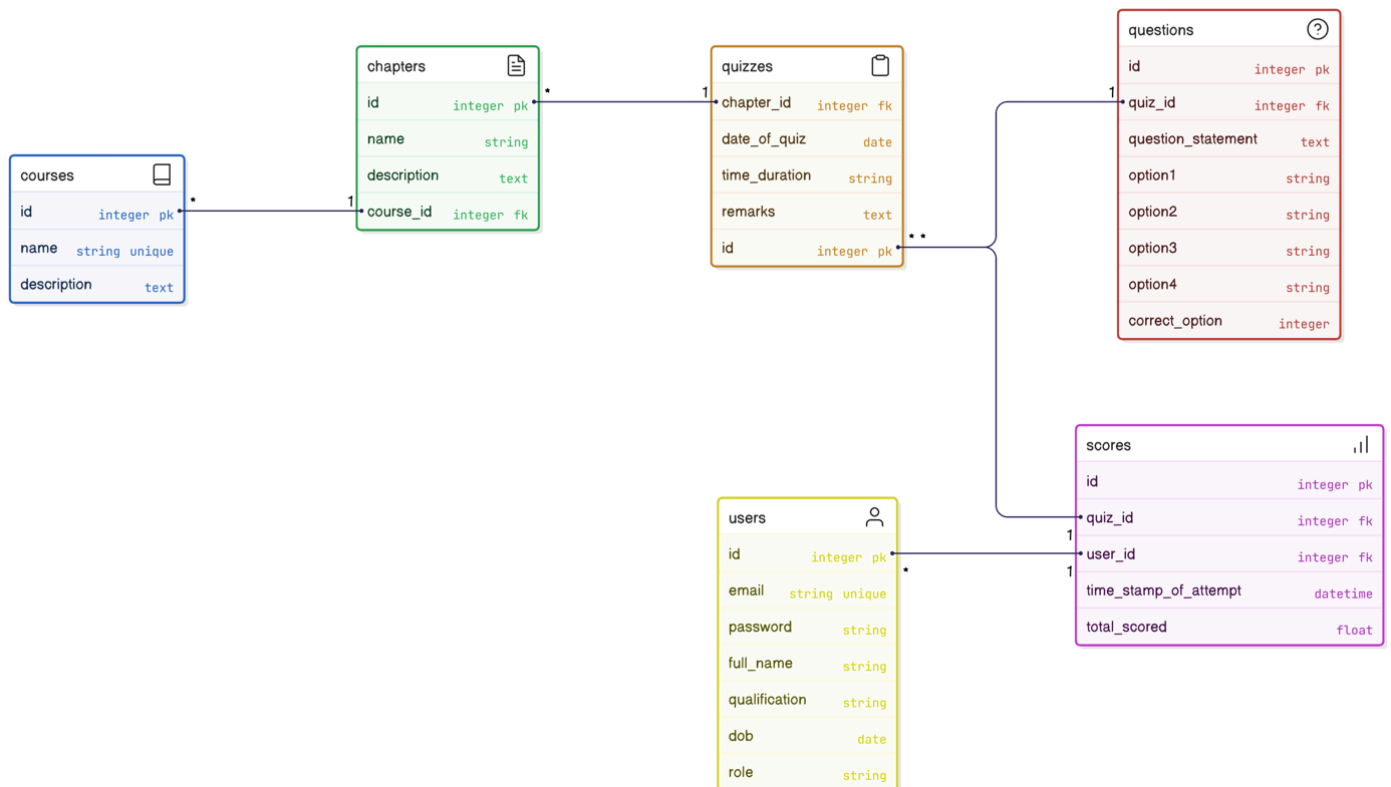
- **Python:** Handles backend logic and API development.
- **Flask-SQLAlchemy:** Manages the database interactions.
- **DateTime:** Tracks the date and time of quizzes.
- **VueJS:** Builds the frontend interface and fetches data from the API.
- **JWT (JSON Web Token):** Provides user data security, preventing unauthorized access.
- **Celery:** Implements background tasks like sending emails and repetitive actions.
- **Redis:** Used for caching to enhance performance.

Architecture and Features:

- **Log-in/Sign-up System:** Users register with email, password, and personal details. Admin does not require registration and has a pre-configured account.
- **Profile Page:** Users can view their quiz history and scores
- **Admin Dashboard:** Admin can create, edit, or delete subjects, chapters, quizzes, and questions.
- **Quiz Management:** Admin can specify quiz details such as duration, date, and MCQ questions. Quizzes are structured under subjects and chapters.
- **Quiz Attempt System:** Users can attempt quizzes, which are timed and automatically scored. Scores are stored for future reference.

- **Daily Quiz Reminders:** The system sends automated reminders via email for users who haven't attempted quizzes recently.
- **Monthly Activity Report:** Users receive a detailed report via email summarizing their quiz activity.
- **Caching and Performance Optimization:** Redis is used for caching frequent database queries, improving response times.

ER Diagram



Video Demonstration Link:

<https://drive.google.com/file/d/1jHttGxdIALCXWavURujZ-8usDmrbaYrx/view?usp=sharing>